GROUNDTRUTH ARCHITECTURE STANDARD

Ensuring Text-Based Invariants

How to guarantee that agent interpretation never deviates away from source ground truth.

RELIABILITY STANDARD
Zero Hallucination

THE GOLDEN PRINCIPLE OF GROUND TRUTH

"The principle is to capture the essentials of the text so that interpretation will never deviate away from the ground truth."

Downstream LLM interpretation naturally drifts when unconstrained. Extracting the irreducible invariant essentials creates an unbreakable tether.

The Polite Suggestion (Prompt Only)

Telling an LLM: *"Please don't approve loans > $150 or forget the hardware key."*

Why it fails: Jailbreaks, context window drift, or persuasive users easily bypass prompt instructions. It's a cardboard sign, not a lock.

The Electric Fence (Text-Based Invariant)

Compiling the text into **structural DAG forks & interceptors**.

Why it succeeds: The execution runtime physically blocks invalid state mutations before dispatch. Even if the LLM hallucinating tries, the fence refuses to open.

FOUR ARCHITECTURAL WAYS TO ENSURE INVARIANTS

1. CONTRACT EXTRACTION

5-Tier Policy Tree

Parse raw text into 5 immutable layers: Root Invariants, Scoped Directives, Rules, Bounds, and Dead Ends.

```
Tier 0: Root Invariant
Tier 3: Bounds [0, $150]
Tier 4: Dead-End Abort
```

2. CODE INTERCEPTORS

Zero-LLM Gatekeeper

Place deterministic Python/JS code before every tool execution. Never let an LLM self-police its own bounds.

```
if amount > 150.00:
    abort_to_tripwire()
if ttl > 14400: revoke()
```

3. ZERO IMAGINED NUMBERS

Respect Algebraic Limits

If text states algebraic bounds without physical constants, keep them algebraic [u_min, u_max]. Never invent 42.0 kPa.

```
True Bounds: Stated Only
Unspecified: Algebraic
Invented: 0% Allowed
```

4. CIRCUIT BREAKER DAG

Hard Terminal Tripwires

Wire invariants directly into LangGraph decision diamonds. Any breach instantly transitions to safe state reversion.

```
Diamond: Limit OK?
No → 🛑 TRIPWIRE
Yes → ✅ Safe Action
```

RUNTIME ENFORCEMENT: THE CLOSED-LOOP INVARIANT GUARDRAIL

STEP 1: INGESTION
Raw Text / URL
SOP, Statute, Novel

→

STEP 2: COMPILER
Extract Invariants
Axioms & Bounds

→

DECISION DIAMOND
Invariant Checked?
Deterministic Code

→

BREACH / DEVIATION
🛑 **Safe Reversion**
Instant Freeze & Alert

OR →

100% SATISFIED
✅ **Action Dispatched**
Faithful Ground Truth

HOW IT ENFORCES GROUND TRUTH ACROSS DOMAINS

DOMAIN	TEXT INVARIANT	DOWNSTREAM RISK IF LEFT TO LLM	ENFORCEMENT CIRCUIT BREAKER
IT SECURITY **SOC2 / Break-Glass**	Physical FIDO2 key mandatory; SMS OTP strictly prohibited; Write lease capped at 4h.	Agent "politely" offers SMS fallback during user login friction or leaves lease open indefinitely.	**Hard Token Shield:** Missing WebAuthn challenge or TTL > 4h triggers automated SIRT audit and lease termination.
FINTECH AML **Dispute Intake**	Federal Non-Tipping Invariant (31 U.S.C. § 5318(g)): Prohibited from disclosing SAR freezes.	Chatbot answers customer: *"Your funds are held due to suspicious activity investigations."*	**Anti-Tipping Redaction Gate:** Pattern interceptor catches SAR keywords; auto-routes dispute to 72h frozen queue.
LITERATURE **Little House / Gatsby**	Pa's discomfort with crowded Wisconsin vs. going West to open Indian country where grass is thick.	Agent inverts polarity or invents corporate IT telemetry for Laura Ingalls Wilder's frontier narrative.	**Semantic Polarity Router:** Red Stagnation maps staying in Wisconsin; Green Action maps departing West.